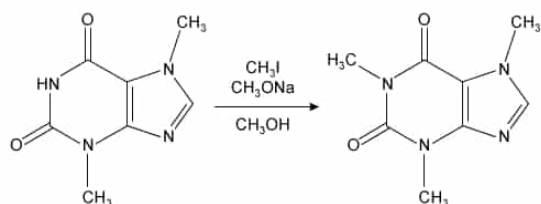


Consider the reaction below.



Which of the following observations would most likely indicate that the product from this reaction is pure?

- ☐ A. One absorbance peak from size-exclusion chromatography [20%]
- ☐ B. Two peaks in a high-performance liquid chromatogram [11%]
- ☐ C. Two unique $-\text{CH}_3$ signals in the ^1H NMR spectrum [17%]
- ☒ D. A single spot in the reaction mixture lane on a thin-layer chromatography plate [51%]

Correct

50% answered correctly

Explanation:

TLC analysis of a reaction mixture

The diagram illustrates the TLC analysis of a reaction mixture. It shows the chemical structures of theobromine (1,3-dimethylxanthine) and caffeine (1,3,7-trimethylxanthine). The reaction conditions are CH_3I , CH_3ONa , and CH_3OH . The TLC process is shown in four steps: 1. Starting material (theobromine) and reaction mixture are spotted on the TLC plate. 2. The TLC plate is placed in a solvent chamber. 3. Sample components are separated as the solvent front moves up the plate. 4. A single product spot (more nonpolar than theobromine) indicates the product (caffeine) is pure.

TLC = thin-layer chromatography. ©UWorld

Several techniques can be used to evaluate the **purity** of a **compound**, including thin-layer chromatography (**TLC**), which separates compounds in a mixture based on **polarity**. In TLC, the tested sample is dissolved in an organic solvent and a small amount is spotted onto the stationary phase (commonly a polar silica [SiO_2]) coated on a TLC plate. After spotting, the TLC plate is placed in a solvent chamber, and the mobile phase (organic solvent) travels up the plate and carries the spotted compounds along at different rates, which separates the mixture components. Nonpolar compounds have less affinity for a polar stationary phase and will travel further up the plate than polar compounds.

The conversion of theobromine to caffeine results in *N*-methylation (conversion of N-H to N-CH_3), resulting in a **large change in polarity**. The N-H in theobromine can **hydrogen bond** to the polar stationary phase on the TLC plate, whereas the N-CH_3 in caffeine **cannot hydrogen bond**. Because theobromine can hydrogen bond, it is more polar and has a smaller **R_f value**

than caffeine, and it produces distinguishably different spots on a TLC plate. Therefore, TLC can be used to assess product purity by comparing a standard of pure theobromine to the isolated product. If a **single spot** with a larger R_f than that of theobromine is visible on the reaction mixture lane on the TLC plate, this indicates that the reaction proceeded to completion, no more reactant remained, and the **isolated product is pure**.

(Choice A) Size-exclusion chromatography separates compounds by size; therefore, compounds that are similar in size elute together. Theobromine and caffeine are very similar in size, differing only by a methylene group. Size-exclusion chromatography is unlikely to provide enough resolution to distinguish the two molecules; instead, they are likely to elute together and appear in one peak.

(Choice B) Although high-performance liquid chromatography (HPLC) can be used to determine product purity, a pure reaction product would show only one peak (not two) on the HPLC chromatogram.

(Choice C) There are *three* unique methyl ($-\text{CH}_3$) groups in the reaction product caffeine; therefore, three unique $-\text{CH}_3$ signals would be seen in its ^1H NMR spectrum. Two unique $-\text{CH}_3$ signals would indicate the product had not formed because the reactant (ie, theobromine) has only two unique $-\text{CH}_3$ groups.

Educational objective:

Thin-layer chromatography (TLC) is a technique used to evaluate a compound's purity. The components of a mixture are separated based on polarity, and a single spot on a TLC plate is indicative of a compound's purity.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 401691

System:
Separation Techniques, Spectroscopy, and Analytical Methods